

How to append vendor specific IE in driver management frames

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Introduction

This document can teach user, how to append vendor specific IE in Realtek driver management frames by iwpriv (rtwpriv) command.

Please follow those steps to set vendor ie setting.

Driver setting.

1. Please enable append vendor ie feature from Makefile
CONFIG_APPEND_VENDOR_IE_ENABLE = n ==>
CONFIG_APPEND_VENDOR_IE_ENABLE = y

Step.

1. Insert Realtek driver
Ex: insmod 8821au.ko
2. Up interface
Ex: ifconfig wlan0 up
3. Use iwpriv tools (or rtwpriv tools) and command `vendor_ie_set` to set vendor ie
Ex: iwpriv wlan0 vendor_ie_set
0,5,dd3300e04c0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef
4. Can use iwpriv tools (or rtwpriv tools) and command `get` to get vendor ie setting
5. Run hostapd or wpa_supplicant

Note: If want to change vendor IE , after setting , must restart hostapd or wpa_supplicant.

Setting example: To set group 0 or group 3

0,5,dd3300e04c0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef
0123456789abcdef

3,3,dd3300e04c0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef
0123456789abcdef

2. vendor_ie_get:

iwpriv [interface_name] vendor_ie_get [vendor_ie_number]

interface_name: wifi interface name

vendor_ie_number: want check appended vendor ie number , can support number 0~4 , totals 5 groups

vendor ie.

ex: To get group 0 setting or get group 3 setting

```
iwpriv wlan0 vendor_ie_get 0
wlan0    vendor_ie_get:
Vendor IE num 0 , Mask:5 [Beacon][Probe Resp]
Vendor IE:
dd3300e04c0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef01234
56789abcdef
or
iwpriv wlan0 vendor_ie_get 3
wlan0    vendor_ie_get:
Vendor IE num 3 , Mask:3 [Beacon][Probe Req]
Vendor IE:
dd3300e04c0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef01234
56789abcdef
```

3. Clear vendor ie setting:

ex: To clear vendor ie number 1 group

```
iwpriv wlan0 vendor_ie_set 1,0,0
```

Can use vendor_ie_get to check

```
iwpriv wlan0 vendor_ie_get 0
wlan0    vendor_ie_get:
Vendor IE num 0 , Mask:0
Vendor IE:
```